

PROJECT BLUEPRINT & ARCHITECTURE

Next-Gen AI-Enabled Real Estate CRM

Custom Solution for Kohinoor Group

Prepared For:	Kohinoor Group Executive Leadership Team
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Target Domain:	High-Velocity Residential Flat Sales & Lifecycle Management
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1. Executive Technical Summary & Tech Stack

To support the scale of Kohinoor Group's premium residential pipeline, a decoupled, asynchronous, API-first architecture is mandatory. **Yes, FastAPI is the highly recommended framework for this backend.** Real estate operations rely heavily on concurrent live inventory edits, background webhooks from lead channels (Facebook/MagicBricks), and dynamic heavy calls to Large Language Models (LLMs) via WhatsApp APIs. FastAPI's native asynchronous capabilities (async/await) mean it can process thousands of concurrent connections using an exceptionally low server resource footprint compared to traditional architectures like Django or Flask.

Recommended Enterprise Tech Stack

Layer	Technology Chosen	Architectural Justification
Backend Framework	FastAPI (Python 3.11+)	Asynchronous execution speeds up LLM/WhatsApp API integrations; auto-generates interactive Swagger/OpenAPI docs for internal apps.
Frontend App	Next.js (React) & Tailwind CSS	Server-Side Rendering (SSR) ensures optimal speed for management dashboards and interactive tower matrices.
Primary Database	PostgreSQL	ACID compliance ensures complex payment tables and flat transactional lock safety never duplicate bookings.
Caching & Locks	Redis	Coordinates high-speed 45-minute flat-parking states across distributed site-sales environments.
AI & NLP Integration	LangChain / OpenAI API (or Private Llama 3 via vLLM)	Powers real-time site-agent conversational companions and automatic customer document extraction (OCR).
Task Queue	Celery & RabbitMQ	Handles detached workflows like batch generation of construction milestone demand letters and WhatsApp push schedules.

2. End-to-End Application Feature Flow

The system organizes the customer and transaction journey through six seamless modules:

- **Phase 1: Inbound & Omni-Channel AI Nurturing:** Leads arrive via aggregators (99acres, Housing.com) or Meta Ads. The integrated **FastAPI Webhook** triggers a secure, low-latency call to the AI WhatsApp

engine. An instant AI interaction validates basic client requirements (BHK preferences, operational budget) before human routing occurs.

- **Phase 2: Intelligent Routing & Lead Scoring:** The lead passes through a custom-built predictive scoring model. High-scoring leads get flagged immediately and pushed via customizable logic (e.g., Round Robin) straight to the mobile dashboard of dedicated site-sales managers.
- **Phase 3: Interactive Floor Wheel & Token Parking:** During physical site visits, the sales rep opens the **Visual Tower Stack Matrix**. If a customer finalizes a flat, the rep initiates a 45-minute temporary flat park. This safely locks the database row via Redis, preventing any duplicate bookings while tokens clear.
- **Phase 4: MahaRERA Onboarding & OCR Compliance:** Upon receipt of the initial token, the system invites the customer to submit official records (Aadhaar, PAN). An OCR pipeline extracts data fields instantly to populate legal agreements, validating mandatory MahaRERA timeline checklists automatically.
- **Phase 5: Construction-Linked Accounting (CLP):** As on-site project operations log physical completions (e.g., Slab 5 casted), FastAPI triggers background tasks via Celery to compute individual balances, draft formal localized demand notes, and track incoming bank clearances.
- **Phase 6: Channel Partner Hub & Post-Sales Handover:** External Channel Partners access an isolated portal to claim their commissions and review certified inventory. Simultaneously, buyers move into a 'Sada Sukhi Raho' customer journey dashboard tracking construction updates and scheduling key handover phases.

3. Functional Feature Breakdown

A. AI WhatsApp Conversation & Qualification Engine

- Continuous 24/7 autonomous engagement via authorized WhatsApp Cloud APIs.
- Instant file sharing of pricing sheets and multi-angle floor maps based on contextual intent parsing.
- One-click programmatic synchronization of qualified buyers into human schedules.

B. Dynamic Tower Grid Inventory Controller

- Complete color-coded UI charting Tower status (Green: Available, Yellow: Parked, Red: Booked).
- Distributed Redis Lock layer ensuring zero-conflict multi-user session management.
- Dynamic pricing controller calculating floor-rise premiums and premium view additions dynamically.

C. Construction-Linked Accounts & Smart Billing

- Structural milestone mapping connecting real-world engineering statuses straight to invoicing flows.
- Secure payment gateway hookups (Razorpay/Stripe/NEFT tracking) enabling split token payment logging.
- Automated delivery schedules for standard statutory statements of accounts and interest warnings.

D. Executive Analytics Suite

- Dynamic Average Selling Price (ASP) trackers charting developer margins across operational quarters.
- Marketing attribution matrices showing real-time customer acquisition costs (CAC) per project tier.
- Predictive cancellation alerts indicating structural pipeline risks before they manifest.

4. Phases of Implementation & Project Timeline

The system is structured as an iterative 24-week deployment roadmap:

Phase	Duration	Core Target Deliverables
Phase 1: Discovery & Database Schema Design	Weeks 1 - 4	Data schema design mapping towers, flats, and leads. Setting up core PostgreSQL tables and FastAPI foundational framework routing layouts.
Phase 2: Sales Engine & Live Inventory Grid	Weeks 5 - 10	Deployment of Next.js frontend, interactive 3D tower stack visual charts, and Redis distributed row locking infrastructure.
Phase 3: Omnichannel AI & Bot Automation	Weeks 11 - 14	Integration of WhatsApp Cloud API, setup of LangChain agent pipelines, and automated OCR processing engines for legal onboarding documentation.

Phase 4: CLP Accounts & ERP Sync Gateway	Weeks 15 - 18	Creation of automated construction-linked billing flows, PDF generation modules for demand notes, and SAP data connection webhooks.
Phase 5: Channel Partner Portal & Analytics	Weeks 19 - 22	Deployment of secondary dashboard for outside brokers and development of real-time multi-dimensional executive revenue intelligence visuals.
Phase 6: UAT, Penetration Testing & Deployment	Weeks 23 - 24	Rigorous multi-site user acceptance testing (UAT), stress testing concurrent unit bookings, final deployment on AWS Cloud systems.